



GC1015

SAMPLING, BIAS, CONFIDENCE INTERVALS, AND PRECISION

Quantitative Methods for Business Analysis



Use what you already know

RECALL

Questions from Session 3 and Session 4. Answer from memory: no notes, no slides.

1

From Session 3: Distribution Center, Spread, and Sample Variance

- Why do we divide by $n - 1$ rather than n when calculating sample variance?

2

From Session 4: Probability Rules and Event Independence

- What mathematical condition must hold for two operational events A and B to be statistically independent?

3

How to answer

- Write one sentence for each question in 3 minutes.
- **Mark how sure you are:** 1 = guess, 2 = fairly sure, 3 = certain. The next slides check your answers.

CHOOSE, THEN EXPLAIN

You are Cedar Loop's analyst reporting how much pickup times vary in a sample. Choose an option, then briefly explain.

Why do we divide by $n - 1$ rather than n when calculating sample variance?

- ☐ A Dividing by $n - 1$ corrects the variance underestimated when using the sample mean.
- ☐ B Dividing by n makes sample variance unbiased because every pickup is counted.
- ☐ C Dividing by $n - 1$ makes the sample mean equal the population mean.

Pick one, then tell a neighbor why in one sentence.

CHOOSE, THEN EXPLAIN

You are reviewing two events in Cedar Loop's pickup records. Your team needs to describe their relationship.

What mathematical condition must hold for two operational events A and B to be statistically independent?

A $P(A \text{ and } B) \text{ equals } P(A) \text{ times } P(B).$

B $P(A \text{ and } B) \text{ equals zero.}$

C $P(A \text{ and } B) \text{ equals } P(A) \text{ plus } P(B).$

Pick one, then tell a neighbor why in one sentence.



Three steps toward a
decision

TODAY'S INVESTIGATION

Three steps toward a decision

1

Key point

- 1. Sampling Design, Bias, and the Limits of Generalization.
- 2. Sampling Distributions, Standard Error, and Mean Estimation.

2

Key point

- 3. Proportion Estimation, Sample Size Planning, and Decision Guardrails.



Slides hand off to the lab,
the lab hands back

TODAY'S LOOP: SLIDES AND LAB

Every section ends in your notebook and workbook; every practice returns you here

1

Sections before the break

- After each section's retrieval slide, a hand-off slide sends you to the lab for one 10 to 10 minute practice: Excel first, Statistical Package for the Social Sciences (**SPSS**) second, with a slide-case example and a fresh example.

2

Break after Section 2

- Return for Sections 3, then the guided build. Complete the planned applied work.

3

Then the studio and the close

- Group studio (Cedar Loop Fulfillment Staffing Decision Studio), synthesis and exit ticket. Open now: the lab notebook in Colab and the workbook in Excel.



A precise number can still
mislead

WHY THIS MATTERS

A precise number can still mislead

1

Key point

- A sample statistic describes the records observed.
- A confidence interval describes uncertainty from sampling, while an audit of missing records addresses a different concern: whether the observed records fairly represent the pickups of interest.

2

Key point

- Managers need both checks before treating an estimated delay rate as a basis for permanent spending.

DECISION POINT: CEDAR LOOP GROCERY PICKUP

Fulfillment manager's staffing decision



Situation

Cedar Loop Grocery Pickup's fulfillment manager has 2,400 completed pickups across four weeks. Of 125 randomly invited pickup IDs, 100 confirmed; 24 exceeded 50 minutes. The slow-pickup target is at most 20%.

A

Authorize permanent evening staff

Benefit: Responds quickly to possible delays.

Risk: Permanent spending may precede verification.

B

Require a bounded verification pilot

Benefit: Checks missing outcomes before permanent spending.

Risk: Delays may persist during the pilot.

Your decision

Would you authorize permanent evening fulfillment staff now or require a bounded verification pilot first, and what evidence would change your mind?

Make a provisional choice; revisit it after the audit.

SECTION

SAMPLING DESIGN, BIAS, AND THE LIMITS OF GENERALIZATION

Audit sampling frames, contrast probability designs, and compute nonresponse scenario bounds to quantify unmeasured delay risks. Populations, Frames, and Probability Sampling Designs · Diagnosing Selection Bias, Nonresponse, and Independence Violations

TARGET POPULATION AND SAMPLING FRAME

Populations, Frames, and Probability Sampling Designs



Shift Stratum	Eligible Pickups	Frame Share
Morning Shift	1,440	0.60
Evening Shift	960	0.40
Total Frame	2,400	1.00

Assuming a 60/40 morning/evening split, proportional allocation of 125 invitations is 75 morning and 50 evening. These stratum counts are hypothetical, not observed audit totals.

Hypothetical morning/evening split of the 2,400-order Cedar Loop frame | Illustrative four-week planning allocation | Unit: Completed pickups

Populations, Frames, and
Probability Sampling
Designs



Populations, Frames, and
Probability Sampling
Designs

TARGET POPULATION AND SAMPLING FRAME

Populations, Frames, and Probability Sampling Designs

1

Meaning: Core Definitions

- **Target population:** The entire group of interest (all 2,400 completed pickups across the four-week audit window).
- **Parameter:** A fixed, unknown population property (true delay proportion p).

2

Meaning: Core Definitions

- **Statistic:** A value computed from sample data (observed sample proportion $\hat{p}$).

3

The Sampling Frame

- The physical roster used to select cases.
- Must be audited to remove canceled orders, drop duplicates, and prevent systematic coverage errors.

SAMPLE PROPORTION POINT ESTIMATE (1/3)

$$\hat{p} = \frac{x}{n} = \frac{\sum_{i=1}^n X_i}{n}$$

Symbols:

$\hat{p}$: Sample proportion statistic estimating the population parameter p

x : Number of observed sample units meeting the operational criteria (slow pickup)

n : Total number of usable observed sample units

X_i : Binary indicator variable equal to 1 if unit i exceeds 50 minutes, and 0 otherwise

Inputs: Observed count of delayed pickups $x = 24$; usable customer response sample size $n = 100$.

SAMPLE PROPORTION POINT ESTIMATE (2/3)

$$\hat{p} = \frac{x}{n} = \frac{\sum_{i=1}^n X_i}{n}$$

Steps:

1. Identify the number of observed successes (delayed pickups): $x = 24$
2. Identify the total number of usable responses: $n = 100$
3. Divide the count of delayed pickups by total responses: $\hat{p} = 24 / 100$
4. Evaluate quotient: $\hat{p} = 0.24$, representing 24.0%

SAMPLE PROPORTION POINT ESTIMATE (3/3)

$$\hat{p} = \frac{x}{n} = \frac{\sum_{i=1}^n X_i}{n}$$

Result: $\hat{p} = 0.24$ (24.0% delayed pickups)

In the responding sample, 24.0% of pickups exceeded the 50-minute threshold, compared to the 20.0% managerial guardrail target.

Use: Used to compute baseline point estimates for binary business KPIs across audit samples.

Limit: The sample statistic reflects only responding units and does not account for potential nonresponse bias among unconfirmed orders.

EXCEL DRAW: FROZEN KEYS (1/3)

```
=LET(  
    audit_frame, FILTER(Frame!A2:C2401, Frame!A2:A2401<>""),  
    ranked_frame, SORTBY(audit_frame, TAKE(audit_frame,,-1), 1, TAKE(audit_frame,,1), 1),  
    sample_draw, CHOOSEROWS(TAKE(ranked_frame,,2), SEQUENCE(125)),  
    sample_draw  
)
```

Symbols: audit_frame: Three-column eligible frame containing unique order IDs, shift labels, and frozen random keys.

TAKE(audit_frame,,-1): Frozen random-key column used for the primary sort. TAKE(audit_frame,,1): Unique order IDs used as a tie-breaker.

ranked_frame: The complete frame ranked by frozen random key and order ID. TAKE(ranked_frame,,2): ID and shift columns retained for the selected draw. sample_draw: The first 125 rows of that ranked ID/shift array; the final LET argument returns it.

Inputs: On Frame, A2:B2401 contains 2,400 eligible unique completed-pickup order IDs and shift labels. Generate random keys in C2:C2401, then freeze them before entering the displayed formula in AuditDraw!A2.

EXCEL DRAW: FROZEN KEYS (2/3)

```
=LET(  
    audit_frame, FILTER(Frame!A2:C2401, Frame!A2:A2401<>""),  
    ranked_frame, SORTBY(audit_frame, TAKE(audit_frame,, -1), 1, TAKE(audit_frame,, 1), 1),  
    sample_draw, CHOOSEROWS(TAKE(ranked_frame,, 2), SEQUENCE(125)),  
    sample_draw  
)
```

Steps:

1. Validate eligibility, unique order IDs and the 2,400-row frame first.
2. Enter =RAND() in Frame!C2 and fill through C2401. Copy C2:C2401 and Paste Special as Values before selecting the draw; retain a dated copy of the complete keyed frame.
3. Enter the displayed formula in AuditDraw!A2; it spills 125 rows and two columns from the frozen-key ranking.
4. Check =ROWS(A2#) and =ROWS(UNIQUE(TAKE(A2#,, 1))); both must return 125.
5. Copy AuditDraw!A2:B126 and Paste Special as Values. Retain these selected IDs with the frozen keyed frame and selection procedure.

EXCEL DRAW: FROZEN KEYS (3/3)

```
=LET(  
    audit_frame, FILTER(Frame!A2:C2401, Frame!A2:A2401<>""),  
    ranked_frame, SORTBY(audit_frame, TAKE(audit_frame,, -1), 1, TAKE(audit_frame,, 1), 1),  
    sample_draw, CHOOSEROWS(TAKE(ranked_frame,, 2), SEQUENCE(125)),  
    sample_draw  
)
```

Result: A frozen tabular list of exactly 125 unique pickup order IDs selected uniformly without replacement.

Under this simple random selection mechanism over the validated frame, each eligible order has selection probability $125/2,400$ (5.21%). Frozen keys and unique-ID tie-breaking retain the ranking for audit; frozen selected IDs retain the draw.

Use: Operational sampling audits for order fulfillment, fraud screening, and quality inspections.

Limit: RAND keys can change on recalculation until frozen. The frozen frame and selected IDs document this draw; they do not validate response quality or independence of outcomes.

THE MECHANISM OF PROBABILITY SAMPLING

Populations, Frames, and Probability Sampling Designs

Stratum	Population Size	Stratum Weight	Allocated Sample (n=125)
Morning Shift	1,440	0.60	75
Evening Shift	960	0.40	50
Combined Operation	2,400	1.00	125

Assuming a 60/40 morning/evening split, proportional allocation of 125 invitations is 75 morning and 50 evening. These stratum counts are hypothetical, not observed audit totals.

Hypothetical morning/evening split of the 2,400-order Cedar Loop frame | Illustrative four-week planning allocation | Unit: Pickup invitations



Populations, Frames, and
Probability Sampling
Designs

THE MECHANISM OF PROBABILITY SAMPLING

Populations, Frames, and Probability Sampling Designs

1

How it works: Probability Sampling Mechanism

- Every unit in the frame has a known, non-zero inclusion probability.
- Selection is governed by quantifiable chance rather than subjective human convenience.

2

Stratified Allocation Realities

- Sample every stratum; allocation can be proportional or disproportionate.
- Combine estimates using justified population weights; realized precision depends on the design and response.

3

Stratified Allocation Realities

- Random invitations do not remove coverage gaps, measurement flaws, or nonresponse bias.



Diagnosing Selection Bias,
Nonresponse, and
Independence Violations

SELECTION BIAS, NONRESPONSE, AND INDEPENDENCE DEFINED

Diagnosing Selection Bias, Nonresponse, and Independence Violations

1

Meaning

- Core Definitions Selection bias systematically excludes specific operational groups during sampling. Nonresponse bias arises when missing units differ systematically from respondents. Independence requires each record to supply separate information.
- Operational Context At Cedar Loop grocery pickup, busy shoppers facing excessive delays may skip follow-up confirmations.

2

Meaning

- Duplicate customer accounts or shared technical outages cluster order outcomes, creating an illusion of large sample sizes while distorting operational metrics.

NONRESPONSE SENSITIVITY SCENARIO BOUNDS (1/3)

$$p_{\min} = \frac{x_{\text{resp}} + 0}{n_{\text{selected}}}, \quad p_{\max} = \frac{x_{\text{resp}} + m}{n_{\text{selected}}}$$

Symbols:

x_{resp} : number of observed positive outcomes (defects or slow pickups) among responding units

m : count of selected units with missing outcomes (nonrespondents), where $m = n_{\text{(selected)}} - n_{\text{resp}}$

$n_{\text{(selected)}}$: total number of units initially selected and invited in the probability draw

$p_{\min}$: best-case proportion among all selected units, assuming zero nonrespondents had the defect

$p_{\max}$: worst-case proportion among all selected units, assuming all nonrespondents had the defect

NONRESPONSE SENSITIVITY SCENARIO BOUNDS (2/3)

$$p_{\min} = \frac{x_{\text{resp}} + 0}{n_{\text{selected}}}, \quad p_{\max} = \frac{x_{\text{resp}} + m}{n_{\text{selected}}}$$

Inputs: $x_{\text{resp}} = 24$, $m = 25$, $n_{\text{selected}} = 125$, $n_{\text{resp}} = 100$

Steps:

1. Calculate the best-case numerator assuming 0 missing orders were slow: $24 + 0 = 24$.
2. Divide best-case numerator by selected sample size:
 $p_{\min} = 24 / 125 = 0.192$ (19.2%).
3. Calculate the worst-case numerator assuming all 25 missing orders were slow: $24 + 25 = 49$.
4. Divide worst-case numerator by selected sample size:
 $p_{\max} = 49 / 125 = 0.392$ (39.2%).

NONRESPONSE SENSITIVITY SCENARIO BOUNDS (3/3)

$$p_{\min} = \frac{x_{\text{resp}} + 0}{n_{\text{selected}}}, \quad p_{\max} = \frac{x_{\text{resp}} + m}{n_{\text{selected}}}$$

Result: Sensitivity bounds for selected sample = [19.2%, 39.2%]

The proportion of slow orders among the 125 selected invitations must lie between 19.2% and 39.2%, demonstrating that nonresponse risk spans across the 20% managerial guardrail.

Use: Evaluate decision vulnerability to nonresponse in customer satisfaction surveys, operational quality audits, and compliance checks.

Limit: Bounds apply deterministically to the selected sample; they do not calculate sampling variability for the remaining unselected units in the population.

CALCULATING NONRESPONSE SENSITIVITY BOUNDS

Diagnosing Selection Bias, Nonresponse, and Independence Violations

Scenario	Responding Slow	Assumed Missing Slow	Total Slow	Selected Base	Selected Proportion
Best-Case Missing	24	0	24	125	0.192
Observed Respondents	24	N/A	24	100	0.24
Worst-Case Missing	24	25	49	125	0.392

Nonresponse sensitivity bounds span from 19.2% to 39.2%, crossing the 20% management guardrail.

Selected pickup orders with confirmation invitations | Specified four-week period | Unit: Pickup orders



Diagnosing Selection Bias,
Nonresponse, and
Independence Violations

CALCULATING NONRESPONSE SENSITIVITY BOUNDS

Diagnosing Selection Bias, Nonresponse, and Independence Violations

1

Worked example

- Cedar Loop Nonresponse Sensitivity Out of 125 selected orders, 100 responded with 24 slow pickups. Missing data for 25 orders create unmeasured risk across the 20% guardrail.
- Scenario Bounds Calculation Best case (0 slow): $(24 + 0) / 125 = 19.2\%$. Worst case (25 slow): $(24 + 25) / 125 = 39.2\%$.

2

Worked example

- Because the interval $[19.2\%, 39.2\%]$ straddles the 20% guardrail, operational follow-up is necessary.

AI SEARCH & SHARE



Discuss with a partner;
artificial intelligence (AI)
support is optional.

APPLY THE IDEA

Discuss with a partner; artificial intelligence (AI) support is optional.

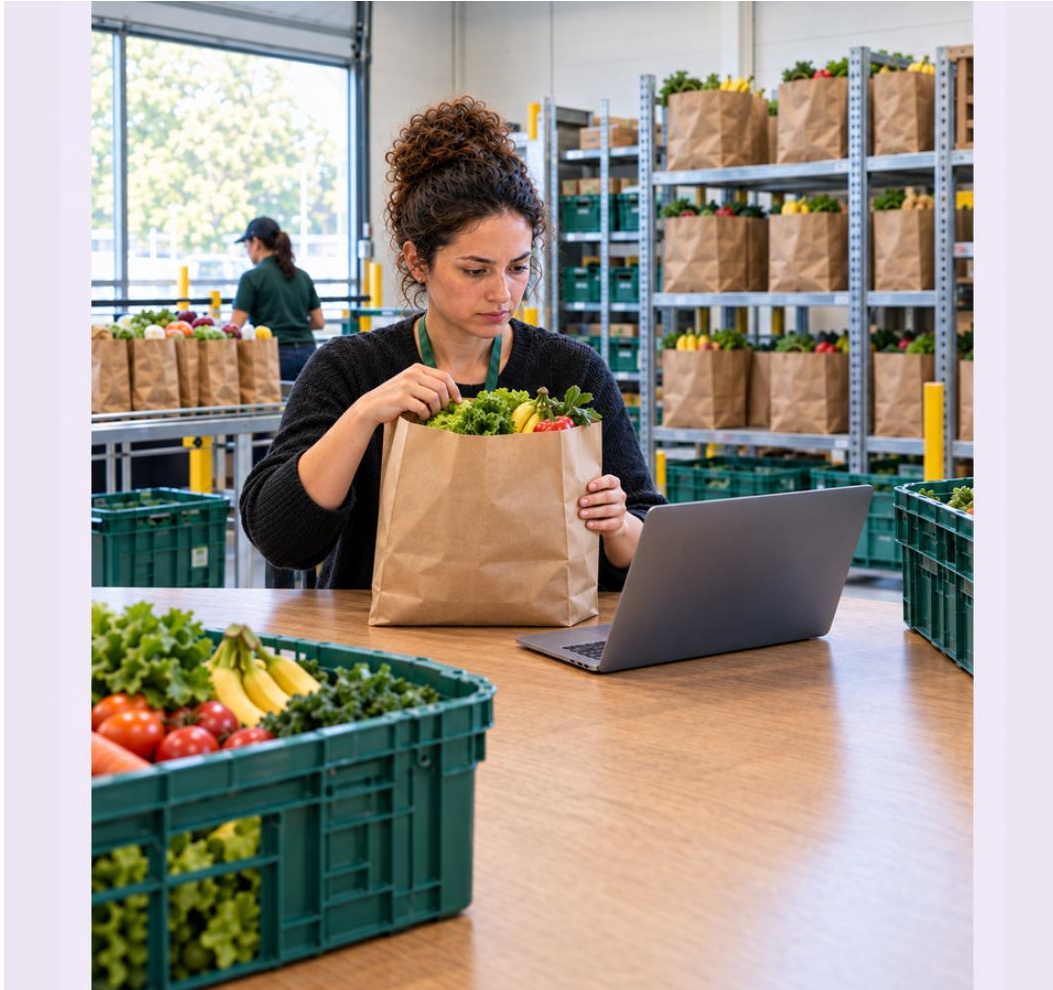
1

Key point

- Can matching respondents to nonrespondents on demographic or operational covariates prove that nonresponse bias is absent?
- Discuss with 1-2 neighbors. Prompt **AI**, check its answer, then volunteer one brief finding when ready.

DECISION POINT: SAMPLING DESIGN, BIAS, AND THE LIMITS OF GENERALIZATION

Use the evidence, explain your decision, and name one limit.



Key Point

Why does an audited simple random invitation of 125 orders fail to guarantee an unbiased final sample when 25 orders fail to confirm?

Name one limit

Sensitivity scenario bounds define deterministic mathematical extremes but do not provide a probability distribution over missing outcomes.

The evidence

A invited: 125; A upper scenario: 39.2%; A eligible: 100.



Excel first, SPSS second

LAB PRACTICE 1: AUDITING CEDAR LOOP SAMPLING BOUNDS

10 minutes in the lab, then return to Section 2 (slide 29)

1

Open and do

- S05P_Learning_Workbook.xlsx > IN-CLASS_Lab Part A: **B7:B12**; interpretation **B14**; reason **B15:H17**.
- Audit invitation, response and observed-outcome denominators; calculate the respondent share and nonresponse bounds. Optional **SPSS**: run complete **H1**, including its filters.

2

Check and bring back

- Bring the bounds and explain whether they cross the management guardrail. Save this workbook. Work 10 minutes; debrief 2 minutes.

3

Help or go further

- **Need help:** use the Part A guide and linked study help.
- **Finish early:** Check the upper bound if all missing outcomes were delayed.

SECTION

SAMPLING DISTRIBUTIONS, STANDARD ERROR, AND MEAN ESTIMATION

Derive sampling distributions using the central limit theorem and construct 95 percent Student t confidence intervals for pickup durations. The Central Limit Theorem and Standard Error of the Mean · One-Sample Confidence Intervals for Continuous Means in Excel and SPSS



The Central Limit Theorem
and Standard Error of the
Mean

WHAT STANDARD ERROR MEANS IN BUSINESS OPERATIONS

The Central Limit Theorem and Standard Error of the Mean

1

Individual Dispersion vs Estimate Stability

- **Standard deviation ($s = 12.00$ min):** Measures unit-to-unit spread among individual pickup handoff durations.
- **Standard error ($SE = 1.20$ min):** Measures the expected variation of the sample mean across repeated samples.

2

Business Implication

- Individual customer experiences remain widely dispersed even when the operational average is precisely estimated.
- Knowing that the average wait is 42 minutes does not prevent individual shoppers from experiencing 65-minute delays.

STANDARD ERROR OF THE MEAN (1/2)

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

Symbols:

SE_(x̄): estimated standard error of the sample mean

s: sample standard deviation of the observed values

n: number of independent usable observations

Inputs: Sample standard deviation $s = 12.00$ minutes,
sample size $n = 100$ usable pickup durations.

Steps:

1. Calculate square root of sample size: $\sqrt{100} = 10$
2. Divide sample standard deviation by square root of n :
 $12.00 / 10 = 1.20$ minutes

STANDARD ERROR OF THE MEAN (2/2)

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

Result: $SE_{\bar{x}} = 1.20$ minutes

Across repeated random samples of 100 pickups, the sample mean duration varies around the population mean with an expected spread of 1.20 minutes.

Use: Used to determine the precision of sample averages and calculate confidence interval margins of error for continuous operational KPIs.

Limit: Assumes observations are independent and identically distributed. Does not correct for nonresponse, measurement bias, or clustered operational delays.

THE MECHANISM OF THE CENTRAL LIMIT THEOREM AND THE SQUARE-ROOT RULE

The Central Limit Theorem and Standard Error of the Mean

Sample Size (n)	Square Root of n	Sample Std Dev (s)	Standard Error (SE)
25	5.0	12.00 min	2.40 min
100	10.0	12.00 min	1.20 min
400	20.0	12.00 min	0.60 min
1600	40.0	12.00 min	0.30 min

Standard error decreases as the square root of sample size increases, demonstrating diminishing marginal precision.

Cedar Loop eligible grocery pickups | Four-week operational window | Unit: Minutes per pickup



The Central Limit Theorem
and Standard Error of the
Mean

THE MECHANISM OF THE CENTRAL LIMIT THEOREM AND THE SQUARE-ROOT RULE

The Central Limit Theorem and Standard Error of the Mean

1

How it works: How the Central Limit Theorem Operates -

- **Shape convergence:** Averages of independent samples approach normality even when individual durations are skewed. - The square-root rule^{**}: **SE** = $s / \sqrt{n}$.
- Dispersion of the mean shrinks inversely with $\sqrt{n}$.^{**}

2

Diminishing Returns in Precision

- **Sample size 100:** **SE** = $12 / 10 = 1.20$ minutes.
- **Sample size 400:** **SE** = $12 / 20 = 0.60$ minutes. Doubling precision requires a fourfold increase in verified sample size.



One-Sample Confidence
Intervals for Continuous
Means in Excel and SPSS

UNDERSTANDING CONFIDENCE INTERVALS FOR OPERATIONAL MEANS

One-Sample Confidence Intervals for Continuous Means in Excel and SPSS

1

Confidence Interval Definition

- A confidence interval bounds an unknown population parameter with a chosen long-run coverage rate.
- Point estimates vary across samples; intervals quantify this estimate-level uncertainty explicitly.

2

Business Example

- Cedar Loop observes 100 pickups with mean 42.00 minutes.
- The interval bounds the population mean duration, not individual customer wait times.

3

Business Example

- A 95 percent level reflects long-run procedure success across repeated audits, not individual transaction spread.

ONE-SAMPLE STUDENT T CONFIDENCE INTERVAL FOR A POPULATION MEAN (1/3)

$$\bar{x} \pm t^* \cdot \frac{s}{\sqrt{n}}$$

Symbols:

$\bar{x}$: Sample mean duration (point estimate of population mean μ)

t^* : Two-tailed critical value from the Student t distribution at significance level α with degrees of freedom $df = n - 1$

s : Sample standard deviation of observed durations

n : Number of usable, independent observations in the sample

$(s/\sqrt{n})$: Estimated standard error of the sample mean (SE)

Inputs: For Cedar Loop grocery pickup: $n = 100$ usable pickups, $\bar{x} = 42.00$ minutes, $s = 12.00$ minutes, confidence level = 95% ($\alpha = 0.05$), $df = 100 - 1 = 99$.

ONE-SAMPLE STUDENT T CONFIDENCE INTERVAL FOR A POPULATION MEAN (2/3)

$$\bar{x} \pm t^* \cdot \frac{s}{\sqrt{n}}$$

Steps:

1. Compute standard error: $SE = s / \sqrt{n} = 12.00 / \sqrt{100} = 12.00 / 10 = 1.2000$ minutes
2. Identify critical value: $t^* = t_{(0.05/2, 99)} \approx 1.9842$ (from Excel =T.INV.2T(0.05, 99))
3. Calculate margin of error: $M = t^* \cdot SE = 1.9842 \cdot 1.2000 = 2.3810$ minutes
4. Calculate lower bound: $\bar{x} - M = 42.00 - 2.3810 = 39.6190 \approx 39.62$ minutes

ONE-SAMPLE STUDENT T CONFIDENCE INTERVAL FOR A POPULATION MEAN (3/3)

$$\bar{x} \pm t^* \cdot \frac{s}{\sqrt{n}}$$

Steps: 5. Calculate upper bound: $\bar{x} + M = 42.00 + 2.3810 = 44.3810 \approx 44.38$ minutes

Result: 95% CI: [39.62, 44.38] minutes

Assuming independent probability sampling and negligible nonresponse bias, this interval procedure provides 95% long-run coverage for the true mean duration of the 2,400 eligible pickups.

Use: Evaluate operational compliance against business KPI targets (for example, verifying whether average duration remains below 40.00 minutes).

Limit: Assumes data are free from extreme measurement errors and that nonrespondents do not differ systematically from respondents.

EXCEL 365 AND SPSS SYNTAX FOR CONTINUOUS MEAN CONFIDENCE INTERVALS (1/3)

* SPSS Syntax for 95% CI on pickup minutes.

```
EXAMINE VARIABLES=minutes  
  /STATISTICS=DESCRIPTIVES  
  /CINTERVAL 95.
```

Excel Formula Reference (assuming summaries in cells E2:E4):

E2 = Sample Size (100)

Symbols: EXAMINE: SPSS command for exploratory data analysis and descriptive inference /STATISTICS=DESCRIPTIVES: Subcommand requesting summary statistics including mean and standard deviation

/CINTERVAL 95: Subcommand setting two-sided confidence level to 95 percent

T.INV.2T: Excel formula returning two-tailed inverse Student t cutoff for probability alpha and degrees of freedom

Inputs: Audited row-level dataset in SPSS containing numeric variable 'minutes'; or Excel summary cells containing count, mean, and standard deviation.

EXCEL 365 AND SPSS SYNTAX FOR CONTINUOUS MEAN CONFIDENCE INTERVALS (2/3)

```
# E3 = Sample Mean (42.00)
# E4 = Sample SD (12.00)
# SE:      =E4/SQRT(E2)
# Critical t: =T.INV.2T(0.05, E2-1)
# Lower Bound: =E3 - T.INV.2T(0.05, E2-1)*(E4/SQRT(E2))
# Upper Bound: =E3 + T.INV.2T(0.05, E2-1)*(E4/SQRT(E2))
```

Steps:

1. In Excel: enter =T.INV.2T(0.05, 99) to obtain critical t-value 1.9842
2. In Excel: enter =E4/SQRT(E2) to obtain standard error 1.2000
3. In Excel: subtract and add product of critical t and standard error to mean
4. In SPSS: execute EXAMINE syntax on active dataset with CINTERVAL 95

EXCEL 365 AND SPSS SYNTAX FOR CONTINUOUS MEAN CONFIDENCE INTERVALS (3/3)

```
# E3 = Sample Mean (42.00)
# E4 = Sample SD (12.00)
# SE:      =E4/SQRT(E2)
# Critical t: =T.INV.2T(0.05, E2-1)
# Lower Bound: =E3 - T.INV.2T(0.05, E2-1)*(E4/SQRT(E2))
# Upper Bound: =E3 + T.INV.2T(0.05, E2-1)*(E4/SQRT(E2))
```

Result: SPSS and Excel output: Lower Bound = 39.62 minutes, Upper Bound = 44.38 minutes

Both software platforms yield identical interval boundaries that bound the true mean duration under frequentist sampling assumptions.

Use: Standardizing reproducible KPI reporting across analytics teams using spreadsheet and statistical software pipelines.

Limit: SPSS syntax requires raw row-level data; Excel summary formulas check arithmetic but cannot independently verify row-level missingness or scanner anomalies.



One-Sample Confidence
Intervals for Continuous
Means in Excel and SPSS

WORKED EVIDENCE: CEDAR LOOP MEAN HANDOFF DURATION

One-Sample Confidence Intervals for Continuous Means in Excel and SPSS

Metric	Calculation and value
Sample and estimate	$n = 100$ usable pickups; mean = 42.00 min
SD and mean SE	$s = 12.00$ min; $SE = 12 / \text{sqrt}(100) = 1.20$ min
Critical t and margin	$df = 99$; $t^* = 1.9842$; $M = 1.9842 * 1.20 = 2.38$ min
95% mean CI	$42.00 \pm 2.38 = [39.62, 44.38]$ min

The interval estimates the population mean under the sampling assumptions; it does not describe individual pickup durations.

Eligible Cedar Loop grocery pickups | Four-week operating audit | Unit: Minutes per pickup



One-Sample Confidence
Intervals for Continuous
Means in Excel and SPSS

WORKED EVIDENCE: CEDAR LOOP MEAN HANDOFF DURATION

One-Sample Confidence Intervals for Continuous Means in Excel and SPSS

1

Worked example: Step-by-Step Cedar Loop Calculation

- **Input summaries:** $n = 100$ pickups, mean = 42.00 min, $s = 12.00$ min, $df = 99$.
- **Standard error:** $SE = 12.00 / \sqrt{100} = 1.2000$ minutes.

2

Worked example: Step-by-Step Cedar Loop Calculation

- **Critical value:** $t^* = 1.9842$ from Excel = **T.INV.2T**(0.05, 99).
- **Margin of error:** $M = 1.9842 * 1.20 = 2.3810$ minutes.

3

Final Interval Bounds

- **Lower Bound:** $42.00 - 2.3810 = 39.62$ minutes.
- **Upper Bound:** $42.00 + 2.3810 = 44.38$ minutes.

AI SEARCH & SHARE



Discuss with a partner;
artificial intelligence (AI)
support is optional.

APPLY THE IDEA

Discuss with a partner; artificial intelligence (AI) support is optional.

1

Key point

- How does an unaddressed 20 percent nonresponse rate affect the validity of a computed 95 percent confidence interval?
- Discuss with 1-2 neighbors. Prompt **AI**, check its answer, then volunteer one brief finding when ready.

DECISION POINT: SAMPLING DISTRIBUTIONS, STANDARD ERROR, AND MEAN ESTIMATION

Use the evidence, explain your decision, and name one limit.



Key Point

If a colleague reports a 95 percent interval of 41.88 to 42.12 minutes by dividing s by 100, what operational mistake occurred?

Name one limit

A confidence interval for the mean bounds average performance across all orders, not individual order variation.

The evidence

$\bar{B}$ mean: 42; B t critical: 1.98; B sample standard deviation: 12.00.



Excel first, SPSS second

LAB PRACTICE 2: ESTIMATING CEDAR LOOP MEAN HANDOFF DURATION

10 minutes in the lab, then back here (Then the break (slide 47))

1

Open and do

- S05P_Learning_Workbook.xlsx > IN-CLASS_Lab Part B: **B7:B12**; interpretation **B14**; reason **B15:H17**.
- Calculate the duration mean, sample **SD**, standard error, critical t and 95% mean interval. Optional **SPSS**: run complete **H2**, including its filters.

2

Check and bring back

- Bring the interval in minutes and compare it with the 40-minute benchmark. Save this workbook. Work 10 minutes; debrief 2 minutes.

3

Help or go further

- **Need help:** use the Part B guide and linked study help.
- **Finish early:** Compare the width of a 99% mean interval.



Pause. Reset. Return ready.



PAUSE BEFORE THE DECISION STUDIO

Keep your provisional choice in mind

Stretch, Hydrate, and Reset

SECTION

PROPORTION ESTIMATION, SAMPLE SIZE PLANNING, AND DECISION GUARDRAILS

Estimate binary proportion confidence intervals, solve for required sample sizes, and synthesize findings into managerial decision guardrails. Confidence Intervals for Binary Operational Proportions · Sample Size Selection and Managerial Decision Framing



Confidence Intervals for
Binary Operational
Proportions

MEANING OF A POPULATION PROPORTION AND POINT ESTIMATE

Confidence Intervals for Binary Operational Proportions

1

Binary Operational KPIs

- Binary performance indicators track whether events occur, coded as 1 for presence and 0 for absence.

2

Sample Point Estimate

- The sample proportion $\hat{p}$ equals observed successes x divided by total sample size n , providing a point estimate of parameter p .



Confidence Intervals for
Binary Operational
Proportions

MEANING OF A POPULATION PROPORTION AND POINT ESTIMATE | CONTINUED 1

Confidence Intervals for Binary Operational Proportions

1

Business Context and Variance Structure

- At Cedar Loop grocery pickup, 24 of 100 confirmed pickups were slow (taking over 50 minutes), yielding $\hat{p} = 0.24$ or 24%.

2

Meaning

- Because variance depends directly on p through p times 1 minus p , precision requires evaluating both successes and failures.

ONE-SAMPLE CONFIDENCE INTERVAL FOR A POPULATION PROPORTION (1/3)

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Symbols:

$\hat{p}$: Sample proportion of successes, equal to x / n

z^* : Two-sided critical value from the standard normal distribution (1.960 for 95% confidence)

n : Number of usable independent observations

x : Count of observed successes in the sample

Inputs: Sample count $x = 24$ slow pickups; usable sample size $n = 100$ pickups; confidence level = 95% ($z^* = 1.960$).

Steps:

1. Calculate point estimate: $\hat{p} = 24 / 100 = 0.24$

ONE-SAMPLE CONFIDENCE INTERVAL FOR A POPULATION PROPORTION (2/3)

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Steps:

2. Check normal approximation: $n\hat{p} = 100(0.24) = 24 \geq 10$ and $n(1 - \hat{p}) = 100(0.76) = 76 \geq 10$
3. Compute standard error: $SE = \sqrt{(0.24)(0.76) / 100} = \sqrt{0.001824} \approx 0.042708$
4. Compute margin of error: $M = 1.960 \times 0.042708 \approx 0.083708$ (8.37 percentage points)
5. Form interval: $[0.24 - 0.083708, 0.24 + 0.083708] = [0.156292, 0.323708]$

Result: 95% CI: $[0.1563, 0.3237]$, or 15.63% to 32.37% of eligible pickups.

ONE-SAMPLE CONFIDENCE INTERVAL FOR A POPULATION PROPORTION (3/3)

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Under repeated random sampling, approximately 95 percent of intervals constructed with this procedure contain the fixed population proportion of slow pickups.

Use: Evaluating whether operational performance meets service-level agreements and binary quality standards.

Limit: Assumes independent random sampling; the Wald interval can underperform when sample size is small or p is close to 0 or 1; does not correct for nonresponse bias.

CONSTRUCTING THE CEDAR LOOP CONFIDENCE INTERVAL

Confidence Intervals for Binary Operational Proportions

Metric	Calculation and value
Sample and delay count	$n = 100$ usable pickups; $x = 24$ pickups > 50 min
Delay share (p-hat)	$24 / 100 = 0.2400$ (24.00%)
Proportion SE	$\text{sqrt}(0.24 * 0.76 / 100) = 0.0427$ (4.27 pp)
Critical z and margin	$z^* = 1.960$; $M = 1.960 * 0.04271 = 0.0837$ (8.37 pp)
95% proportion CI	$0.24 \pm 0.0837 = [15.63\%, 32.37\%]$

This conditional interval estimates the underlying delay share. The margin is 8.37 percentage points.

Eligible Cedar Loop grocery pickups | Four-week operational audit | Unit: Pickups and proportions



Confidence Intervals for
Binary Operational
Proportions

CONSTRUCTING THE CEDAR LOOP CONFIDENCE INTERVAL

Confidence Intervals for Binary Operational Proportions

1

Point Estimate and Standard Error

- Cedar Loop observes 24 slow pickups out of 100 usable confirmations: $\hat{p} = 0.24$; **SE** = $\sqrt{0.24 * 0.76 / 100} = 0.04271$ (4.27 percentage points).

2

Margin of Error Calculation

- At 95% confidence, $z^* = 1.960$. Margin of error $M = 1.960 * 0.04271 = 0.08371$ (8.37 percentage points).

3

95% Confidence Interval and Target Limit

- Adding and subtracting M gives $[0.1563, 0.3237]$, or 15.63% to 32.37% slow pickups. Because this interval straddles the 20% benchmark, the sample cannot prove a performance breach.



Sample Size Selection and
Managerial Decision
Framing

WHAT SAMPLE SIZE PLANNING MEANS IN BUSINESS ANALYTICS

Sample Size Selection and Managerial Decision Framing

1

Conditional planning

- Choose confidence, margin and assumed variability to set a completed-sample target under a justified sampling and response model.

2

Cedar Loop decision

- Compare the delay estimate with the 20 percent service guardrail; also check missing outcomes and dependence.

3

Limits

- Realized precision can differ. More records do not remove bias; invitations do not guarantee completed responses.

SAMPLE SIZE FORMULA FOR A POPULATION PROPORTION (1/4)

$$n = \frac{(z^*)^2 \cdot p \cdot (1 - p)}{E^2}$$

Symbols:

n: Required minimum number of usable, independent observations

z*: Two-sided critical value from the standard normal distribution corresponding to the chosen confidence level (for example, 1.960 for 95 percent)

p: Planning proportion estimating the true population parameter (use 0.50 for conservative planning)

E: Desired margin of error expressed as an absolute decimal proportion (for example, 0.04 for 4 percentage points or 0.05 for 5 percentage points)

Inputs: Confidence level = 95% ($z^* = 1.960$), planning proportion $p = 0.24$, target margin of error $E = 0.04$.

SAMPLE SIZE FORMULA FOR A POPULATION PROPORTION (2/4)

$$n = \frac{(z^*)^2 \cdot p \cdot (1 - p)}{E^2}$$

Steps:

1. Calculate numerator critical square: $(1.960)^2 = 3.8416$
2. Calculate parameter dispersion product: $p * (1 - p) = 0.24 * (1 - 0.24) = 0.24 * 0.76 = 0.1824$
3. Multiply numerator terms: $3.8416 * 0.1824 = 0.70070784$
4. Square target margin of error: $(0.04)^2 = 0.0016$
5. Divide numerator by squared margin: $0.70070784 / 0.0016 = 437.9424$
6. Round up to next whole integer: $n = 438$ usable observations

SAMPLE SIZE FORMULA FOR A POPULATION PROPORTION (3/4)

$$n = \frac{(z^*)^2 \cdot p \cdot (1 - p)}{E^2}$$

Result: n = 438 usable completed pickup records.

Under simple random sampling and valid response assumptions, a sample of at least 438 usable completed pickup records yields a planned margin of error of at most 4.00 percentage points at 95 percent confidence if p is near 0.24. This illustrative 4-point scenario requires substantially more data than the separate five-point planning illustration (281 usable pickups, or 352 invitations at 80 percent response; 385 under p = 0.50).

SAMPLE SIZE FORMULA FOR A POPULATION PROPORTION (4/4)

$$n = \frac{(z^*)^2 \cdot p \cdot (1 - p)}{E^2}$$

Use: Use when scoping operational audits, customer satisfaction surveys, fulfillment accuracy checks, and digital experiment cohort sizes.

Limit: Assumes simple random sampling, independent observations, and negligible nonresponse bias. It is a normal-approximation planning formula that does not eliminate bias or guarantee actual coverage under non-random selection.

WORKED SAMPLE SIZE CALCULATION FOR CEDAR LOOP PICKUP PRECISION

Sample Size Selection and Managerial Decision Framing

Planning Scenario	Planning Proportion	Target Margin (E)	Raw Formula n	Required Usable n
Empirical Baseline (4 pt)	0.24	0.04	437.94	438
Conservative Default (4 pt)	0.50	0.04	600.25	601
Hypothetical Baseline (5 pt)	0.24	0.05	280.28	281
Hypothetical Conservative (5 pt)	0.50	0.05	384.16	385

Conditional normal-approximation targets under assumed p and valid sampling/response. Completions are not invitations; realized precision can differ.

Cedar Loop eligible pickup orders | Illustrative precision plans | Unit: Completed pickups (planning)



Sample Size Selection and
Managerial Decision
Framing

WORKED SAMPLE SIZE CALCULATION FOR CEDAR LOOP PICKUP PRECISION

Sample Size Selection and Managerial Decision Framing

1

Inputs and Formula

- Using 95 percent confidence ($z = 1.960$), preliminary proportion $p = 0.24$, and managerial margin target $E = 0.04$:

2

Worked example

- $n = (1.960^2 * 0.24 * 0.76) / (0.04^2)$.
- Step-by-Step Result

3

Worked example

- $n = (3.8416 * 0.1824) / 0.0016 = 0.7007 / 0.0016 = 437.94$ usable orders. Fractional units cannot be collected, so round up to 438.
- Interpretation and Planning Comparison

WORKED SAMPLE SIZE CALCULATION FOR CEDAR LOOP PICKUP PRECISION | CONTINUED 1

Sample Size Selection and Managerial Decision Framing



Sample Size Selection and
Managerial Decision
Framing

1

Worked example

- With assumed $p = 0.24$ and valid sampling/response assumptions, the four-point planning target is 438 completed pickups; the separate five-point target is 281. Realized interval width can differ.

AI SEARCH & SHARE



Discuss with a partner;
artificial intelligence (AI)
support is optional.

APPLY THE IDEA

Discuss with a partner; artificial intelligence (AI) support is optional.

1

Key point

- How do nonresponse and selection bias affect the interpretation of a calculated sample size?
- Discuss with 1-2 neighbors. Prompt **AI**, check its answer, then volunteer one brief finding when ready.

DECISION POINT: PROPORTION ESTIMATION, SAMPLE SIZE PLANNING, AND DECISION GUARDRAILS

Use the evidence, explain your decision, and name one limit.



Key Point

If a sample proportion is 24 percent but its 95 percent confidence interval spans 15.6 to 32.4 percent, why is committing capital premature?

Name one limit

Statistical precision formulas assume zero selection bias and cannot repair unrepresentative response rates.

The evidence

C eligible: 100; C completed target: 438; C slow: 24.



Excel first, SPSS second

LAB PRACTICE 3: PLANNING CEDAR LOOP PRECISION GUARDRAILS

10 minutes in the lab, then back here (the group studio (slide 70))

1

Open and do

- S05P_Learning_Workbook.xlsx > IN-CLASS_Lab Part C: **B7:B12**; interpretation **B13**; reason **B14:H14**.
- Count eligible delay_flag outcomes, calculate the delay share and interval, then plan completed pickups. Optional **SPSS**: run complete **H3**, including its filters.

2

Check and bring back

- Bring the rounded-up completed-sample target and explain a conditional decision. Save this workbook. Work 10 minutes; debrief 2 minutes.

3

Help or go further

- **Need help:** use the Part C guide and linked study help.
- **Finish early:** At a hypothetical 70% response rate, plan invitations separately from completed pickups.



PRACTICE

Complete Parts A, B, and C in the issued lab guide

CEDAR LOOP LAB PRACTICE | CONTINUED 1

Complete Parts A, B, and C in the issued lab guide



Complete Parts A, B, and
C in the issued lab guide

1

Part C: Proportion CI and Sample Size Planning.

- Use the issued guide and workbook for each task's file, sheet or software location, cell references, output checks, stop point, and next location.



GROUP STUDIO

Make a recommendation that can be challenged



Cedar Loop Fulfillment
Staffing Decision Studio

GROUP STUDIO: CEDAR LOOP FULFILLMENT STAFFING DECISION STUDIO

15 minutes group work; 16 minutes whole-class sharing

1

Roles and file

- Facilitator, Evidence checker, Recorder, Reporter. Open the **PNP-GRP** practice document.

2

Use today's lab

- Cite at least one result from your lab, with its units, and say what it cannot show before you choose.

3

Bring back

- **One decision recommendation (Option A:** immediate hiring vs Option B: bounded audit pilot), one statistical justification using confidence bounds, and one operational condition that would reverse your choice. Then synthesis (slide 74) and the exit ticket in your **PNP-IND** guided notes.



Make a recommendation
that can be challenged

CEDAR LOOP FULFILLMENT STAFFING DECISION STUDIO | CONTINUED 1

Make a recommendation that can be challenged

1

Evidence question

- If a sample proportion is 24 percent but its 95 percent confidence interval spans 15.6 to 32.4 percent, why is committing capital premature?

2

Evidence question

- Present your choice, supporting evidence, one consequence, and what would change your mind. Follow the issued submission requirement and due date.

CHOOSE, THEN EXPLAIN

You are advising Cedar Loop on a permanent staffing investment after reviewing 100 pickups. Managers want your recommendation today.

If a sample proportion is 24 percent but its 95 percent confidence interval spans 15.6 to 32.4 percent, why is committing capital premature?

- ☐ A The interval crosses the 20 percent limit, so a breach remains uncertain.
- ☐ B The 24 percent estimate proves the 20 percent limit was breached.
- ☐ C The 100 pickups guarantee that the population matches the sample.

Pick one, then tell a neighbor why in one sentence.



Save, resume, reflect

CONTINUE FROM YOUR SAVED CHECKPOINT

The Take home portion completes this session's lab

1

Save before leaving

- Use the checkpoint named in your lab's In class route.

2

Resume at home

- Post-Class Reflection and Decision Revisit
- 'IN-CLASS_Lab Part C'!**B15**.

3

Check and submit

- Complete the listed checks and follow the issued submission instructions. Use the catch-up route if needed.

KEY TAKEAWAYS: REVISE THE CEDAR LOOP DECISION

Vote again, then state the condition



Situation

Cedar Loop's 24 slow pickups among 100 confirmations exceed the 20% target, but 25 of 125 selected pickups did not confirm.



A: Authorize permanent evening staff.

Benefit: Respond quickly to possible delays. Risk: Commit to permanent spending before verification.



B: Require a bounded verification pilot.

Benefit: Audit missing outcomes before permanent spending. Risk: Delays may persist during the pilot.



Vote again, then state the
condition

KEY TAKEAWAYS: REVISE THE CEDAR LOOP DECISION | CONTINUED 1

Vote again, then state the condition

1

Your decision

- Would you authorize permanent evening fulfillment staff now or require a bounded verification pilot first, and what evidence would change your mind? Cast a final vote before the debrief.
- The 19.2% to 39.2% missing-outcome bounds and 15.63% to 32.37% conditional population confidence interval both cross the target. A narrower interval cannot repair biased response.



One optional question
remains

THANK YOU, AND LOOKING AHEAD

One optional question remains

1

Key point

- **Today's decision is conditional:** verify the missing outcomes and sampling process before treating the respondent estimate as grounds for permanent spending.

2

Optional question, not homework

- What additional evidence would help determine whether adding evening fulfillment staff actually reduces slow pickups?



Outside the live teaching
route

STUDENT REFERENCES

Outside the live teaching route

1

Key point

- **GC1015** Session 5 distributed slides and issued lab guide: sampling design and nonresponse; standard error and mean intervals; proportion intervals and sample size planning.
- Excel 365 functions and **IBM SPSS** Statistics procedures are referenced where used in the issued practice materials. No textbook is required.

APPENDIX

APPENDIX A: SELECTED STUDY SUMMARIES

Selected examples for further study. Find complete explanations, worked examples and references under the matching topic in the Session Study Toolkit.

Key point

Selected examples for further study. Find complete explanations, worked examples and references under the matching topic in the Session Study Toolkit.



Populations, Frames, and
Probability Sampling
Designs

WORKED SAMPLE DRAW AND INITIAL ESTIMATION

Populations, Frames, and Probability Sampling Designs

1

Worked Example: Cedar Loop Sample

- **Frame:** $N = 2,400$ deduplicated pickups across four weeks.
- **Selection:** Simple random draw of $n = 125$ pickup invitations.

2

Continued

- **Usable responses:** 100 records collected (25 nonrespondents).

3

Sample Point Estimates

- **Slow pickup share:** $p\text{-hat} = 24 / 100 = 0.24$ (24.0%).
- **Mean pickup duration:** $\bar{x} = 42.00$ minutes ($s = 12.00$ minutes).



Diagnosing Selection Bias,
Nonresponse, and
Independence Violations

COMMON ERRORS IN TREATING NONRESPONSE AND SAMPLE SIZE

Diagnosing Selection Bias, Nonresponse, and Independence Violations

1

Misconception: High Response Guarantees Representativeness

- An 80% response rate does not prevent bias. If the 20% nonrespondents experienced extreme delays, the observed metric severely distorts operational reality.

2

Misconception: Large Samples Fix Bias

- Random contact does not ensure random response. Furthermore, unique order IDs do not guarantee statistical independence when transactions share couriers, accounts, or weather events.



One-Sample Confidence
Intervals for Continuous
Means in Excel and SPSS

COMMON FALLACIES IN INTERVAL INTERPRETATION

One-Sample Confidence Intervals for Continuous Means in Excel and SPSS

1

Fallacy 1: Individual Orders vs. Population Mean

- A confidence interval bounds the population mean, not individual customer wait times.
- 95% of individual orders span roughly 18 to 66 minutes, far wider than [39.62, 44.38].

2

Fallacy 2: Post-Calculation Probability Claims

- The true mean is a fixed constant, not a moving random variable.
- The 95% probability describes the long-run procedure, not the already-computed numbers.



Confidence Intervals for
Binary Operational
Proportions

COMMON MISCONCEPTION: UNITS AND POINT ESTIMATE CERTAINTY

Confidence Intervals for Binary Operational Proportions

1

Error 1: Confusing Units

- Saying margin of error is 8% implies 8% of 24% (1.92 percentage points). Always report proportion margins in percentage points.

2

Error 2: Point Estimate Fixation

- Treating $\hat{p} = 24\%$ as proof that the population exceeds the 20% target ignores sampling uncertainty entirely.

3

Correct Practice

- Report the point estimate alongside its margin in percentage points and evaluate interval bounds against management benchmarks.



Sample Size Selection and
Managerial Decision
Framing

COMMON ERRORS IN SAMPLE SIZE ROUNDING AND REPORTING UNITS

Sample Size Selection and Managerial Decision Framing

1

Misconception 1: Standard Rounding

- Rounding down to the nearest whole integer violates precision targets. A calculation of 437.05 must round up to 438 to meet the conditional planning threshold.

2

Misconception 2: Ambiguous Units

- Stating a margin of error as 4 percent confuses relative change with additive width. Stating 4 percentage points clarifies that the interval extends plus or minus 0.04 from the estimate.

3

Executive Clarity

- Always specify absolute percentage points and integer counts of usable completions in formal decision briefs.

SECTION

FURTHER READING

Optional post-session exploration

SPARSE OUTCOME INTERVALS (1/2)

$$\tilde{p} = \frac{x + 2}{n + 4}, \quad \tilde{p} \pm 2\sqrt{\frac{\tilde{p}(1 - \tilde{p})}{n + 4}}$$

Symbols:

x is the observed event count; n is the number audited.
The adjusted proportion is used to construct the interval, not to replace x/n .

Inputs: Hypothetical audit: $x = 2$ timeouts, $n = 25$ queries;
approximate 95% critical value = 2.

Steps:

1. Observed rate: $2/25 = 8\%$. Adjusted proportion: $4/29 = 13.79\%$.

SPARSE OUTCOME INTERVALS (2/2)

$$\tilde{p} = \frac{x + 2}{n + 4}, \quad \tilde{p} \pm 2\sqrt{\frac{\tilde{p}(1 - \tilde{p})}{n + 4}}$$

Steps:

2. Adjusted standard error is approximately 6.40 percentage points; margin is approximately 12.81 percentage points.

Result: Approximate interval: 0.99% to 26.60%.

The data permit a much higher timeout rate than the observed 8%.

Use: Small audits of uncommon binary failures.

Limit: Endpoints are not guaranteed to stay within 0% and 100%; sampling bias remains possible.

CLUSTERS REDUCE PRECISION (1/2)

$$D = 1 + (m - 1)\rho, \quad n_{\text{eff}} = \frac{n}{D}, \quad SE_{\text{cluster}} = SE_{\text{independent}} \sqrt{D}$$

Symbols:

D is the design effect; m is equal cluster size; rho is within-cluster correlation.

n is the record count; effective sample size expresses equivalent independent information.

Inputs: Hypothetical audit: n = 120 calls, m = 6 calls per agent, assumed rho = 0.25, independent-record SE = 2.00 minutes.

Steps:

1. $D = 1 + (6 - 1)(0.25) = 2.25$; effective sample size = $120/2.25 = 53.3$.

CLUSTERS REDUCE PRECISION (2/2)

$$D = 1 + (m - 1)\rho, \quad n_{\text{eff}} = \frac{n}{D}, \quad SE_{\text{cluster}} = SE_{\text{independent}} \sqrt{D}$$

Steps:

2. Cluster-adjusted SE = $2.00 \times \text{sqrt}(2.25) = 3.00$ minutes.

Result: The assumed clustering raises standard error from 2.00 to 3.00 minutes.

The 120 calls provide precision comparable to roughly 53 independent calls under these assumptions.

Use: Audits with records grouped by agent, route or shift.

Limit: The simple formula assumes approximately equal cluster sizes and a suitable within-cluster correlation; it does not correct bias.

SECTION

INDUSTRY APPLICATION

Optional post-session reading - outside live class time



WHAT DID THE AUDIT SAMPLE?

Claims are the units; dollars are the outcome

1

Organization and frame.

- **HHS-OIG** audited Methodist Hospital's Medicare fee-for-service claims in selected risk areas for January 1, 2020 through December 31, 2021. Its findings concern that defined claims population, not every hospital claim.

2

Measure and evidence.

- In a stratified random sample of 100 claims, auditors found 27 claims with billing errors and \$256,926 in net overpayments. Net overpayment is measured in dollars;
- the count of erroneous claims is a different measure. Because strata and claim amounts affect extrapolation, neither 27/100 nor the sampled dollars can simply be multiplied into a population total.

3

Measure and evidence.

- **Source:** [**HHS-OIG** report A-09-23-03001](<https://oig.hhs.gov/oas/reports/region9/92303001.pdf>).



WHY RECOVER THE LOWER BOUND?

An uncertainty-aware dollar decision

1

Reported estimate.

- For the defined Methodist Hospital audit population and period, **HHS-OIG** reported an estimated \$18,766,379 in net Medicare overpayments. Its two-sided 90% confidence interval ran from \$12,499,714 to \$25,033,044.
- These are extrapolated dollars, not percentages of erroneous claims.

2

Decision calculation.

- **Point estimate minus lower limit:** $\$18,766,379 - \$12,499,714 = \$6,266,665$. Recommending recovery at the lower limit rather than the point estimate leaves that amount of estimated overpayment outside the recommendation.
- The limit reflects sampling uncertainty; it does not establish the exact overpayment. The report's stratified extrapolation cannot be reproduced by multiplying the sample total by a single factor.

3

Decision calculation.

- **Source:** [**HHS-OIG** report A-09-23-03001](<https://oig.hhs.gov/oas/reports/region9/92303001.pdf>).